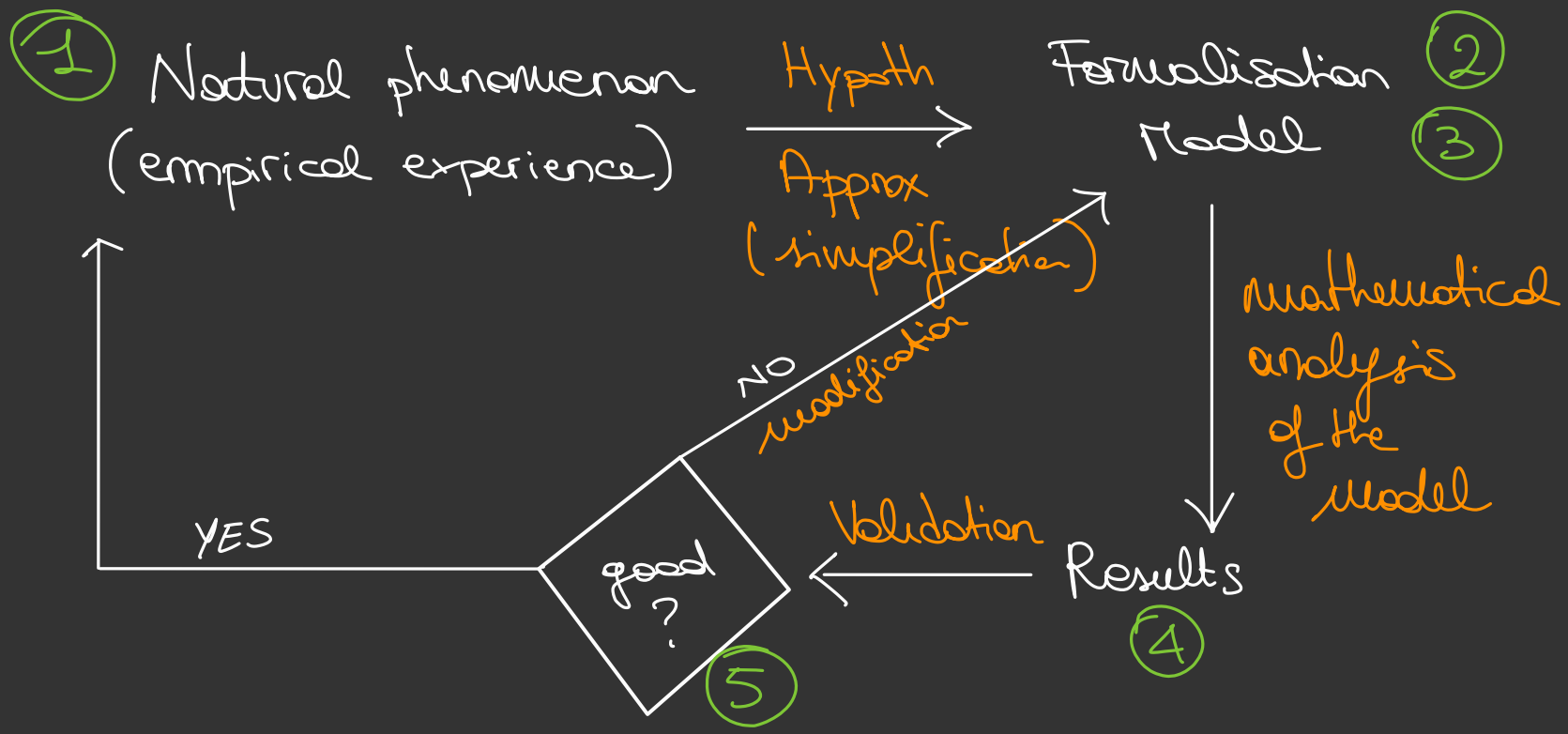
CLASS : Thur - 05.03.2026



Q1 Exist

Q2 regul.

Q3 uniq

Q4 qualitat

Q5 long time behav.
(equi converg asymptotic)

1 Empirical observation of the phenomenon

* main mechanisms
* experimental data (From the lab, statistically, etc.)
* try to understand the dynamics in order to formalise mathematically the main relevant part (and to neglect the less relevant one).

2 Construction of the model

fixed the unknowns and the parameters of your future model.

the variable of the unknowns (typically time (space))

the parameters are the arbitrary constants of the model.

Assume hypothesis and simplifications or/and approximation (starting/toy model)

(3) Formalisation of the model

Choice of the formalism:
* deterministic description (i)
* probabilistic description (ii)

(i) you can describe precisely the state of your system. You know the initial condition and you can describe how it evolves

(ii) you don't know precisely the state (i.e. initial state) of your system
So you use a probability description of the state of the system in a time interval.

Independent variable ⇝ typically t (time)

* discrete $t \in \{1, 2, 3, 4, \dots, t_0\}$.
* continuous $t \in (t_0, t_1)$.

(4) You analyse the model mathematically with the aim to answer to natural questions using math tools (analytical or numerics)
⇝ results

5 Validation of your results
(with respect to initial data or in comparison with empirical experience)

* IF good ⟹ you have a model for your phenomenon that allows you to describe, explain, predict.

* IF not good ⟹ you have to modify your model (Assumption or simple)

We will now describe a scenario when a natural phenomenon is modelled by a mathematical formalism.

Chemical Reaction

Ludwig Wilhelmy (1850)

Described how sugar decomposed into fructose and glucose in presence of an acid

Z sugar

S acid

$\rightsquigarrow \quad \frac{d}{dt} Z = -MZS$

t time variable

* S not supposed to change
* M is a constant depending on the variation of the temperature

today called REACTION VELOCITY CONSTANT

$\rightsquigarrow \quad \ln Z = -M\,S(t-t_0) - \ln Z_0$

$\ln \frac{Z}{Z_0} = -M\,S(t-t_0) \rightsquigarrow \boxed{Z(t) = Z_0 e^{-MS(t-t_0)}}$

Validated with the experiments.

* t independent variable $\rightsquigarrow$ ODE
* x independent variable

$\Big\} \rightsquigarrow$ PDE

(If you assume that the concentration of the substance depends on the spatial distribution)

* linear differential equation :

transition (autonomous transition of the substance into a different element)

(change of the state)

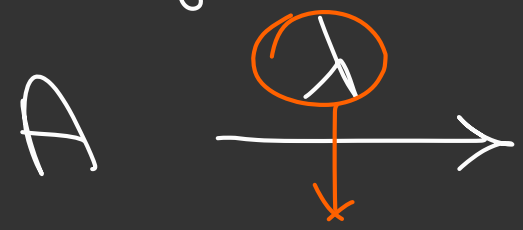
* non linear differential equation :

interactions

→ Law of mass action :

The rate of chemical reaction is directly proportional to the product of the concentrations of the reactants.

(1) Transformation chemical reaction of 1 species



* A, B species of ≠ types: (atoms, molecules, chemical mixture)

* transf. native species

HP1

P(1° time a particle of type A transforms into a particle of type B)

* exponential law with prob. density

$$p(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$P(\Delta t) = \int_0^{\Delta t} \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_0^{\Delta t}$$

prob di trasformazione in intervallo Δt

$$= 1 - e^{-\lambda \Delta t} \quad \Delta t \ll 1 \quad \uparrow \quad \simeq \lambda \Delta t + o((\Delta t)^2)$$

oss

$$\mathbb{P}(\Delta t) = 1 - e^{-\lambda \Delta t} \xrightarrow{\lambda \to +\infty} 1.$$

$\rightsquigarrow$ Binomial prob. laws of transformation

$$\mathbb{P}(\Delta t) = \begin{cases} \lambda \Delta t & \text{transformation} \\ 1 - \lambda \Delta t & \text{non transformation} \end{cases}$$

$$A_{end} - A_{start} \approx - \lambda \Delta t A$$

For the first examples, we first find the explicit solution and then, we study the long time behaviour.

$a(t) = a$:= density of concentration of species A

$b(t) = b$:= " " " B

$$\begin{cases} \dot{a} = -\lambda a, & a(0) = a_0 \\ \dot{b} = \lambda a, & b(0) = 0 \end{cases}$$

NOTATION:

$$\dot{z} = \frac{dz}{dt}$$

$$a(t) = a_0 e^{-\lambda t}$$

$$\rightsquigarrow \quad \dot{b} = \lambda a_0 e^{-\lambda t}$$

$$b(t) = -a_0 e^{-\lambda s}\Big|_0^t$$

$$= a_0 \left(1 - e^{-\lambda t}\right)$$

2) Dissociation

(i) $A \xrightarrow{\lambda} B + C$

$$\begin{cases} \dot{a} = -\lambda a \\ \dot{b} = \lambda a \\ \dot{c} = \lambda a \end{cases} \qquad w := (a, b, c)$$

$$w_0 = (a_0, 0, 0) = w(0)$$

$$\Rightarrow \quad \dot{w} = \Lambda w \,, \text{ with } \quad \Lambda := \begin{bmatrix} -\lambda & 0 & 0 \\ \lambda & 0 & 0 \\ \lambda & 0 & 0 \end{bmatrix}$$

$$a + b \equiv \text{cost}_1$$
$$a + c \equiv \text{cost}_2$$

$$\begin{cases} a(t) = a_0 e^{-\lambda t} \\ b(t) = a_0 \left(1 - e^{-\lambda t}\right) \\ c(t) = a_0 \left(1 - e^{-\lambda t}\right) \end{cases}$$

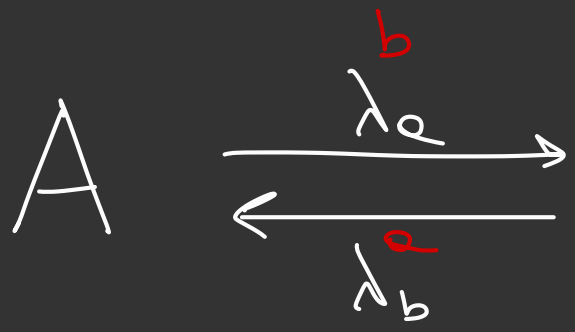
(ii) $A \xrightarrow{\lambda} 2B$

$$\begin{cases} \dot{a} = -\lambda a \\ \dot{b} = 2\lambda a \end{cases}, \qquad \begin{aligned} a(0) &= a_0 > 0 \\ b(0) &= 0 \end{aligned}$$

$$\Rightarrow \boxed{\begin{aligned} a(t) &= a_0 e^{-\lambda t} \\ b(t) &= \int_0^t 2\lambda a_0 e^{-\lambda s}\, ds = 2a_0\left(1 - e^{-\lambda t}\right) \end{aligned}}$$

4) Reversible transition

$\lambda_a^b = \lambda_{a \to b}$



$, \lambda_a, \lambda_b > 0$

$$\begin{cases} \dot{a} = -\lambda_a a + \lambda_b b, & a(0) = a_0 > 0 \\ \dot{b} = \lambda_a a - \lambda_b b, & b(0) = b_0 > 0 \end{cases}$$

$c := a + b \quad \Rightarrow \quad \dot{c} = 0$

$d := \lambda_a a - \lambda_b b \qquad (\dot{d} = \lambda_a \dot{a} - \lambda_b \dot{b})$

$$\Rightarrow \dot{d} = -\lambda_a^2 a + \lambda_a \lambda_b b - \lambda_a \lambda_b a + \lambda_b^2 b$$

$$= -\lambda_a(\lambda_a a - \lambda_b b) - \lambda_b(\lambda_a a - \lambda_b b) = -(\lambda_a + \lambda_b) D$$

$$\begin{cases} \dot{c} = 0 \\ \dot{d} = -(\lambda_a + \lambda_b)\, d \end{cases}, \qquad \begin{aligned} c(0) &= a_0 + b_0 \\ d(0) &= \lambda_a a_0 - \lambda_b b_0 \end{aligned}$$

$$a(t) = \frac{1}{\lambda_a + \lambda_b}\left[a_0\left(\lambda_b + \lambda_a e^{-(\lambda_a+\lambda_b)t}\right) + b_0\lambda_b\left(1 - e^{-(\lambda_a+\lambda_b)t}\right)\right]$$

$$b(t) = \frac{1}{\lambda_a + \lambda_b}\left[a_0\lambda_a\left(1 - e^{-(\lambda_a+\lambda_b)t}\right) + b_0\left(\lambda_a + \lambda_b e^{-(\lambda_a+\lambda_b)t}\right)\right]$$

$$\big(a(t), b(t)\big) \xrightarrow{t \to +\infty} \left(\frac{\lambda_b}{\lambda_a+\lambda_b}(a_0+b_0), \frac{\lambda_a}{\lambda_a+\lambda_b}(a_0+b_0)\right)$$

!!

$$(\bar{a}, \bar{b})$$

Exponential speed of convergence with rate: $\lambda_a + \lambda_b$

$(\bar{a}, \bar{b}) \in \mathbb{R}^2_+$ is the ! sol of $\begin{cases} -\lambda_a a + \lambda_b b = 0 \\ a + b = a_0 + b_0 \end{cases}$

Inep:

In order to reach the equi is sufficient that one of the 2 kinetic constants is larghe enough.

If we consider a perturbation of the initial condition of the equilibrium $(\bar{a}, \bar{b})$, i.e.

$$a(0) = a_0 + a_1 > 0, \qquad b(0) = b_0 + b_1 > 0$$

then how does the long time behaviour of the system change?